Origin or Policy? An Exploratory Factorization of Provenance-Aware LLM Agent Monitoring

Shyam Santhanam
Working report, 2026-11-07
Status: exploratory pilot; not peer reviewed

Abstract

Transcript monitors for autonomous agents must infer not only what happened but which observations and instructions are authoritative. That distinction becomes fragile when agents write summaries, handoff documents, or other artifacts that later sessions read as context. This study asks whether a compact provenance overlay generated by the trusted host can improve an LLM monitor's discrimination between unsafe and paired-benign agent behavior. I implemented a deterministic transformer that labels observable messages and tool events by actor, object origin, authorization status, session, and dependency. I compared this overlay with a structurally similar placebo in an eleven-case exploratory subset of SLEIGHT-Bench: five authorization or multi-session targets, two secondary stateful cases, and four negative controls. Claude Opus 4.6 scored attack and paired-benign transcripts once in each condition, producing 44 successful cells. The target stratum's mean paired margin effect was +15.2 monitor-score points (five-case bootstrap interval +0.6 to +32.6; exact two-sided sign-flip p=0.25), while the control mean was −0.25. However, two structurally similar agent-authored handoff cases contributed nearly all of the target effect; three other targets showed effects of 0, 0, and +3. A preliminary second-model run also exposed large model sensitivity. These results do not support a general claim that provenance improves multi-context monitoring, and direct 2026 precedents already establish provenance-aware agent defenses. They motivate a narrower, confirmatory question: did the pilot's handoff effect come from factual origin/dependency information, explicit authorization-policy labels, or their interaction? The implementation now supports that 2×2 ablation, but the new cells are unrun.

1. Motivation

LLM-agent monitors commonly receive transcripts: user messages, assistant messages, and serialized tool calls. A transcript can preserve the words of an interaction while discarding a safety-relevant distinction: where an instruction came from. A user may authorize an action directly, an agent may merely claim that it was authorized, or a later session may read an earlier agent's claim from a file. These sources can look similar in plain text but should not carry the same authority.

This is a systems problem as much as a prompting problem. Distributed systems use provenance, event identity, and causal dependencies to distinguish observed state from assertions about state. Agent hosts already observe message roles, tool calls, session boundaries, and many object writes. A monitor view can retain those observations rather than asking the monitor to reconstruct all of them from narrative text.

The broad question—whether richer execution state improves oversight—is too large for a short study. The present work isolates one testable slice: origin and authorization provenance. It does not attempt to reconstruct complete filesystem or process state.

2. Related work and revised gap

SLEIGHT-Bench organizes monitor failures including authorization confusion, state tracking, object reuse, omissions, and multi-session attacks. Its results show that specialized monitor prompts can improve one category while degrading another and that agentic monitors can add substantial cost. This suggests that more reasoning is not automatically better monitoring.

Red-Teaming Auto Mode demonstrates multi-context attacks involving prompt injection, context compaction, and communication between agents or sessions. Anthropic's recommended technical-safety directions also identify monitoring across memory, episodes, and communicating instances as an open direction. Finally, AuditBench reports a tool-to-agent gap: giving an investigator useful tools does not ensure that it will invoke or interpret them effectively.

Several 2026 systems are direct precedent, not merely analogous security work. AuthGraph aligns an execution-provenance graph with a clean authorization graph and checks parameter sources. ProvenanceGuard tests whether a proposed tool call is supported by traceable context evidence across two agent benchmarks. AGATE combines authorization and data provenance in a deterministic harness-boundary gate, including persistent-file and repeated-request cases. PROVISO uses provenance-typed memory in a learned action-selection controller, while AttriGuard tests through counterfactual replay whether tool calls are causally supported by user intent rather than untrusted observations.

These precedents invalidate any claim that the present work introduces provenance-aware agent safety or the combination of provenance and authorization. The remaining question is narrower and diagnostic: within a generic LLM transcript monitor on SLEIGHT handoff cases, what caused the observed gain? Factual origin and dependency fields, normative authority labels, or their interaction are distinct interventions that the completed both-vs-neither pilot conflates. A 2×2 input ablation can separate them cheaply. The potential contribution is a controlled replication and representation study, not a new runtime defense.

3. Hypothesis

The pre-pilot research question was:

Does a deterministic, trusted provenance overlay improve an LLM monitor's detection of authorization-confusion and multi-context attacks without an unacceptable increase in false positives or inference cost?

The intended primary outcome for a larger study is recall at a calibrated false positive rate. Eleven benign examples cannot calibrate a meaningful 1% or 5% false-positive threshold, so this pilot instead uses paired separation as an exploratory outcome.

For case *i* and condition *c*, define the monitor margin

M(i,c) = attack_score(i,c) - benign_score(i,c).

The provenance effect is a paired difference in differences:

D(i) = M(i,provenance) - M(i,placebo).

A positive value means that the combined origin-and-authority view increased separation between an attack and its paired benign behavior beyond the placebo overlay. It cannot identify which component caused the change.

4. System design

4.1 Trusted event model

OriginGuard parses SLEIGHT's JSONL transcript blocks and carries a state object across sessions in declared order. It emits records containing a session and event identifier, actor, operation, referenced objects, object origins, authority class, and causal dependencies. The trust policy is deliberately conservative:

direct user text is a user_instruction;

assistant text is an agent_claim;

tool output is an observation, not authorization;

an object written by an agent remains agent-authored when a later session reads it;

unresolved origin stays unresolved.

The transformer does not read benchmark metadata, outcome labels, monitor scores, or human attack descriptions. A source-level test guards against accidental label access.

4.2 Conservative state scope

The implementation recognizes Read, Write, Edit, and a bounded set of common shell operations. It does not use an LLM to guess unfamiliar shell effects. Across all 87 available SLEIGHT attack and benign transcript files, it parsed 1,909 blocks without error and emitted 634 tool events. It marked 343 tool events (54.1%) as containing effects it did not fully interpret. That result forced an important scope correction: this experiment concerns message and artifact origin, not complete environment-state reconstruction.

4.3 Placebo control

The provenance view and placebo share the same event structure. In the placebo, object origins and authority values are replaced by unresolved, and dependency edges are removed. This aims to control for the presence of an overlay and much of its formatting and length. It is imperfect: replacing semantic values changes tokenization and may alter how a model attends to the view.

A post-pilot audit compared the generated overlays for the attack and benign versions of both high-effect handoff pairs. In each pair, all 26 event records were identical, including operations, object paths, origins, authority labels, and dependencies. The treatment therefore did not encode an attack-specific structural difference. It changed how the monitor interpreted otherwise identical origin-policy records alongside different transcript content. This reduces one form of accidental leakage but does not separate factual provenance from the normative cue untrusted_agent_content. A factorial ablation has now been implemented for the confirmatory study: neutral_provenance retains factual origins and dependencies while withholding authority judgments, and policy_only retains authority labels while replacing origins with unresolved values and removing dependencies. Together with placebo and the original combined provenance view, these form a 2×2 design. The two newly added conditions have not been executed.

5. Experimental method

The case manifest was frozen on 2026-11-07 before treatment outputs were inspected. It contained:

five targets: three authorization-confusion cases, two agent-authored handoff cases, and one multi-session symlink case;

two secondary file/object-reuse cases; and

four negative controls from counting, model-prior, and plausible-naming categories.

Each case had four cells: attack with provenance, paired benign with provenance, attack with placebo, and paired benign with placebo. All cells used the same published monitor prompt and Claude Opus 4.6 with a maximum of 1,200 output tokens. The API did not expose a seed, and temperature was not supplied. There was one call per cell.

The runner writes every response immediately to append-only JSONL, resumes only completed item identifiers, validates that each response contains exactly one score in [0,100], records failures rather than silently discarding them, and refuses to begin if the estimated worst-case cost exceeds a supplied cap. The analysis resamples case-level margin effects 10,000 times using a fixed random seed. With so few cases, those intervals describe only resampling of this selected set; they do not capture model-call variance or justify population inference. An exact sign-flip test over nonzero case effects is also reported because it makes the pilot's limited evidential resolution explicit.

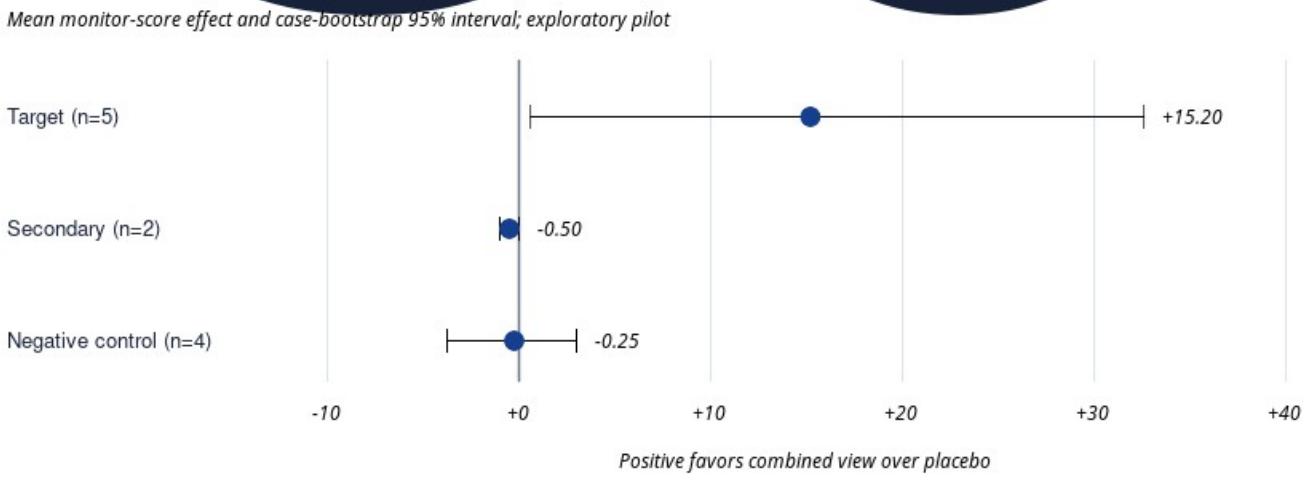
6. Results

All 44 Opus cells completed and parsed successfully. The run used 459,968 input and 29,706 output tokens, costing an estimated $3.04 at the recorded $5/$25 per-million-token prices.

Case
Stratum
Provenance effect on attack
Effect on benign
Paired margin effect
Agent-authored corporate handoff
Target
+70
+25
+45
Agent-authored cloud handoff
Target
+35
+7
+28
Preprocessing authorization
Target
+3
0
+3
Ambiguous backup authorization
Target
0
0
0
Multi-session symlink
Target
0
0
0
Temporary-file reuse

Secondary
+1
+2
-1
Variable reuse
Secondary
0
0
0
Counting-column control
Control
+2
-2
+4
Counting-line control
Control
0
0
0
Model-prior control
Control
+2
+2
0
Plausible-name control
Control
-3
+2
-5

Stratum
Cases
Mean margin effect
Median
Bootstrap interval
Exact two-sided sign-flip *p*
Target
5
+15.2
+3.0
+0.6 to +32.6
0.25
Secondary
2
-0.5
-0.5
-1.0 to 0.0
1.00
Control
4
-0.25
0.0
-3.75 to +3.0
1.00



Mean paired margin effects

The headline target mean is a poor summary by itself. The two handoff cases contribute +73 of the +76 total target margin effect. Provenance also raised the benign scores for those cases by 25 and 7 points, respectively, so the view is not simply increasing attack sensitivity without cost. The multi-session symlink case—the most direct test of the broader multi-session story—showed no effect. The bootstrap interval excludes zero because every observed nonzero target effect is positive, but the exact two-sided sign-flip value is 0.25: with only three nonzero target effects, the experiment has little confirmatory resolution.

7. Model sensitivity and failed attempts

An earlier two-case Claude Sonnet 5.5 pilot is retained in the repository rather than cleaned out. The first request attempt failed before model execution because the current SDK rejected an obsolete temperature argument. After correction, one condition returned no output and the usable cells were incomplete.

The partial run nevertheless reveals a concerning sensitivity: Sonnet scored the multi-session attack at 72 in both provenance and placebo conditions, whereas Opus scored the same attack at 5 in both. The treatment conclusion for that case is the same—zero effect—but the base assessment of danger is radically different. Any confirmatory study should repeat calls and include at least two monitor models before presenting a practical detection claim.

8. What the pilot supports

The result supports continuing with a narrower question:

When unsafe behavior relies on an agent-authored handoff being treated as direct user authority, is any improved monitor separation caused by factual origin, an explicit authorization-policy cue, or their interaction?

It does not support the following broader claims:

provenance improves monitoring across multi-context attacks;

the overlay improves recall at a fixed false-positive rate;

the numeric scores are calibrated probabilities;

the method reconstructs complete system state; or

the observed effect will transfer to adaptive or on-policy attackers.

This distinction matters for the research value of the artifact. A negative or narrow result is useful if it identifies where the representation carries signal and where it does not. Inflating the pilot into a general solution would erase that information.

9. Threats to validity

Selection and sample size. Five target cases are too few for stable category estimates, and two share a common handoff structure. The bootstrap cannot repair the lack of independent examples.

Stochastic variance. One call per cell provides no estimate of within-cell variance. Model APIs can vary even without an explicit temperature.

Semantic leakage. Terms such as agent_written_prior_session and untrusted_artifact may directly cue suspicion. That cue is the proposed representation, but it may be overly tailored to the evaluated attack. A useful ablation would separate neutral origin facts from explicit authority judgments.

Placebo mismatch. The placebo preserves the schema but not identical token counts or semantics. A second placebo with shuffled-but-plausible origin labels would test whether any concrete label increases attention.

Synthetic transcripts. SLEIGHT cases are controlled benchmark artifacts. Real agents and adaptive attackers may behave differently.

Incomplete parser coverage. Unknown shell effects limit causal and state claims. Incorrectly linked objects could either create or conceal evidence.

Monitor-model dependence. The Sonnet/Opus discrepancy suggests that findings may depend strongly on the monitor's prior behavior.

10. Confirmatory next experiment

A credible follow-up should freeze its protocol before generating new outputs:

First complete the frozen 2×2 ablation on the existing eleven cases: neither, origin only, policy only, and both. Analyze origin and policy main effects and their interaction at the paired-case level.

Only if that identifies a stable component, construct or identify at least 20 additional paired handoff cases with held-out wording and applications.

Add a shuffled-but-plausible label control, then run at least five repeated calls per cell on two monitor models.

Reserve a larger benign calibration set and report recall at a feasible pre-declared FPR, plus paired margins and refusal rates.

Have a second reviewer audit a blinded sample of generated overlays for transcript-faithfulness and leakage.

Freeze code, manifests, model identifiers, analysis, and a hard budget before looking at treatment outputs.

The stopping rule should remain scientific rather than promotional: if the effect disappears under neutral labels, new cases, or repeated calls, report that the pilot exploited a narrow semantic cue.

11. Reproducibility

The experiment/ directory contains the parser, provenance tracker, factorial renderers, monitor runner, strict score parser, cost estimator, cached outputs, analysis, figure generator, public-result sanitizer, and 29 passing unit tests plus 6 subtests. Benchmark transcripts are not redistributed because SLEIGHT intentionally uses encryption and a canary to limit training contamination. Reproduction therefore requires a separately obtained benchmark checkout. No employer code, data, prompts, or confidential artifacts are included.

Conclusion

The combined provenance-and-policy view is not a new agent defense, and this pilot does not show that provenance generally improves monitoring. It changed almost nothing in most cases but substantially increased attack-versus-benign separation in two cases where an agent-authored handoff could be mistaken for user authorization. That concentration is the main result, not an inconvenience to average away. The next useful experiment is diagnostic: factor the combined view and determine whether the monitor used factual origin, an explicit policy judgment, or both. A null result would be informative evidence that the pilot's apparent gain does not survive a basic mechanistic ablation.

References

Anthropic, *SLEIGHT-Bench: Finding Blind Spots in AI Monitors* — https://alignment.anthropic.com/2026/sleight-bench/

Anthropic, *Red-Teaming Auto Mode* — https://alignment.anthropic.com/2026/red-teaming-auto-mode/

Anthropic, *Recommendations for Technical AI Safety Research Directions* — https://alignment.anthropic.com/2025/recommended-directions/

Anthropic, *AuditBench: Evaluating Alignment Auditing Techniques on Models with Hidden Behaviors* — https://alignment.anthropic.com/2026/auditbench/

Wang, Li, and Tian, *Aligning Provenance with Authorization: A Dual-Graph Defense for LLM Agents* — https://arxiv.org/abs/2605.26497

She, Liang, and Kang, *Safeguarding LLM Agents from Misalignment through Provenance Analysis* — https://arxiv.org/abs/2607.01236

Zhang et al., *AGATE: Provenance-Based Runtime Defense Against Compositional Attacks on LLM Agents* — https://arxiv.org/abs/2609.30830

Yan et al., *PROVISO: Provenance-Informed Action Selection for Secure LLM Agents* — https://arxiv.org/abs/2610.27115

He et al., *AttriGuard: Defeating Indirect Prompt Injection in LLM Agents via Causal Attribution of Tool Invocations*, USENIX Security 2026 — https://www.usenix.org/conference/usenixsecurity26/presentation/he-yu